

Question 1 _MODULE 5 Assignment

rohit45@rohit45-VirtualBox:~\$ cd dir1

rohit45@rohit45-VirtualBox:~/dir1\$ ls

dir2 f1.txt f2.txt f3.sh f4.csv f.txt smlinkfile.txt

rohit45@rohit45-VirtualBox:~/dir1\$ cat smlinkfile.txt

Original File @ /dir2

rohit45@rohit45-VirtualBox:~/dir1\$ cd dir2

rohit45@rohit45-VirtualBox:~/dir1/dir2\$ ls

orgfile.txt

rohit45@rohit45-VirtualBox:~/dir1/dir2\$ cat orgfile.txt

Original File @ /dir2

rohit45@rohit45-VirtualBox:~/dir1/dir2\$ cd ~

rohit45@rohit45-VirtualBox:~\$ cat file_analyzer.sh

#!/bin/bash

FOUND_GLOBAL=0

rec_filesearch()

{

dir="\$1"

keyword="\$2"

error_flag=0

echo "PROCESSING RECURSIVE SEARCH TO FIND \$keyword in \$dir"

for i in "\$dir"/*;

do

echo "PROCESSING RECURSIVE SEARCH TO FIND \$keyword in \$dir"

if [[-f "\$i"]];then

echo "IN FILE \$i"

```

grep -E "$keyword" "$i"
if [[ "$?" -eq 0 ]];then
#      echo "-----"
      FOUND_GLOBAL=1
      echo "KEYWORD FINDING SUCCESS"
      echo "PATH : $i"
      echo "-----"
      echo
fi
elif [[ -d "$i" ]];then
#      echo "IN SUBDIR $i"
      rec_filesearch "$i" "$keyword"
elif [[ -s "$i" ]];then
      echo "ENTERED AN EMPTY FILE:$i"
fi
done
}

```

```

logger(){
    echo -e "$(date)""\n$1\n" >> errors.log
}

```

here_doc()

```
{
```

```
    cat <<EOF
```

MENU USING HERE DOCUMENT

```
-----
```

In Script : \$0

```
-----
```

Operations allowed :

-d <dir_name> for directory

-f <file_name> for file

-k <keyword> for Keyword Search

--help Display Help Menu

EOF

}

here_str()

{

echo "-----"

echo "IN HERE_STR FN WITH INPUT FILE "\$1" AND KEYWORD "\$2""

if [[! -f "\$1"]];then

logger "File not found in here_str:\$1"

exit 1

else

grep -E "\$2" <<<"\$(cat "\$1")"

if [["\$?" -ne 0]];then

**logger "KEYWORD \$2 IS NOT AVAILABLE IN \$1 (FROM
HERE_STR)"**

fi

fi

echo "-----"

}

validate_ipkeyword()

{

if [[! "\$1" =~ ^[[:alnum:]]+\$]];then

logger "Improper Keyword Entry : "\$1""

exit 1

```
        fi
    }

    if [[ "$1" == "--help" ]];then
        here_doc
        exit 0
    fi

    while getopts ":d:f:k:" opt;
    do
        case "$opt" in
            d)direc=$OPTARG;;
            f)file=$OPTARG;;
            k)keyw=$OPTARG;;
            ?)logger "INVALID OPTION ENTERED IN GETOPTS $OPTARG"; exit 1;;
            :)logger "NO ARG FOR GETOPTS $OPTARG"; exit 1;;
        esac
    done

    echo "SCRIPT: $0"
    echo "TOTAL ARGS : $# "
    echo "ARGS ENTERED : $@"
    echo

    if [[ -z "$keyw" ]];then
        logger "KEYWORD NOT PROVIDED"
        exit 1
    fi
```

```
validate_ipkeyword "$keyw"
```

```
if [[ -n "$direc" ]];then
```

```
    rec_filesearch "$direc" "$keyw"
```

```
    if [[ "$FOUND_GLOBAL" -eq 0 ]]; then
```

```
        logger "Keyword $keyw not found in Recursive search"
```

```
    fi
```

```
elif [[ -n "$file" ]];then
```

```
    here_str "$file" "$keyw"
```

```
else
```

```
    logger "NO VALID INPUT PROVIDED"
```

```
    here_doc
```

```
    exit 1
```

```
fi
```

```
#validate_ipkeyword "@#123"
```

```
#rec_filesearch "dir1" "File"
```

```
#here_str "dir1/dir2/orgfile.txt" "File"
```

```
rohit45@rohit45-VirtualBox:~$ ./file_analyzer.sh --help
```

```
MENU USING HERE DOCUMENT
```

```
-----
```

```
In Script : ./file_analyzer.sh
```

```
-----
```

```
Operations allowed :
```

```
-d <dir_name> for directory
```

```
-f <file_name> for file
```

```
-k <keyword> for Keyword Search
```

```
--help Display Help Menu
```

```
rohit45@rohit45-VirtualBox:~$ ./file_analyzer.sh -d dir1 -k File
```

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -d dir1 -k File

PROCESSING RECURSIVE SEARCH TO FIND File in dir1

PROCESSING RECURSIVE SEARCH TO FIND File in dir1/dir2

Original File @ /dir2

KEYWORD FINDING SUCCESS

PATH : dir1/dir2/orgfile.txt

Original File @ /dir2

KEYWORD FINDING SUCCESS

PATH : dir1/slinkfile.txt

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -d dir1 -k MEPCO

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -d dir1 -k MEPCO

PROCESSING RECURSIVE SEARCH TO FIND MEPCO in dir1

PROCESSING RECURSIVE SEARCH TO FIND MEPCO in dir1/dir2

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -f dir1/dir2/orgfile.txt -k File

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -f dir1/dir2/orgfile.txt -k File

IN HERE_STR FN WITH INPUT FILE dir1/dir2/orgfile.txt AND KEYWORD File

Original File @ /dir2

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -f dir1/dir2/orgfile.txt -k MEPCO

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -f dir1/dir2/orgfile.txt -k MEPCO

IN HERE_STR FN WITH INPUT FILE dir1/dir2/orgfile.txt AND KEYWORD MEPCO

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -d dir1 -k @#123

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -d dir1 -k @#123

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -f wrong.txt -k File

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 4

ARGS ENTERED : -f wrong.txt -k File

IN HERE_STR FN WITH INPUT FILE wrong.txt AND KEYWORD File

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh -d dir1

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 2

ARGS ENTERED : -d dir1

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

Thu Feb 5 11:08:30 PM IST 2026

KEYWORD NOT PROVIDED

rohit45@rohit45-VirtualBox:~\$./file_analyzer.sh

SCRIPT: ./file_analyzer.sh

TOTAL ARGS : 0

ARGS ENTERED :

rohit45@rohit45-VirtualBox:~\$ cat errors.log

Thu Feb 5 11:06:26 PM IST 2026

Keyword MEPCO not found in Recursive search

Thu Feb 5 11:07:15 PM IST 2026

KEYWORD MEPCO IS NOT AVAILABLE IN dir1/dir2/orgfile.txt (FROM HERE_STR)

Thu Feb 5 11:07:40 PM IST 2026

Improper Keyword Entry : @#123

Thu Feb 5 11:07:54 PM IST 2026

File not found in here_str:wrong.txt

Thu Feb 5 11:08:30 PM IST 2026

KEYWORD NOT PROVIDED

Thu Feb 5 11:08:43 PM IST 2026

KEYWORD NOT PROVIDED